Diagrammatic Monte Carlo Computational Lab

Interactions and Correlations in Condensed Matter

Cesare Franchini, Thomas Hahn, Marco Barducci

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1 Setup

1. **Where**: AULA INFORMATICA ZOOLOGIA, second floor, via F. Selmi 3. The PC Lab Room is equipped with Windows PC.
   Optionally you can use your own laptop (Linux suggested).
   Login to Lab PC.
   Username: your complete unibo email address

2. **Connection to the cluster**: Simulations will be physically performed in the OPH cluster.
   The cluster is accessible from the Lab PCs and laptops using SSH.
   Type in the terminal window (Linux terminal or Windows Powershell) the following command:

   ```
   ssh -J name.surname00@bastion-nav.difa.unibo.it name.surname00@ophfe1
   ```

   name.surname00 = your unibo studio account (e.g. cesare.franchini2)
   password: your default Unibo password

   mind: the password is requested twice

   This will do a two-step connection, first to bastion-nav.difa.unibo.it, then to ophfe1.

   To see images and plots from the remote server, avoid disconnections see further details about `ssh` at the end of the guide (section 7).

3. **Setup working directory**: Copy the lab directory in your login directory (/home/STUDENTI/-name.surname00):

   ```
   cp -r /home/STUDENTI/marco.barducci2/DIAGMC-LAB .
   ```

   The directory contains:

   - `Day 1`: Monte Carlo π, Markov Chain Monte Carlo.
   - `Day 2`: DiagMC free propagator, potential and two level system.
   - `Day 3`: Froehlich polaron
   - `env_setup` : setup environment to run the job

Day 1: the codes use python
Day 2-3: the codes run diagrammatic Monte Carlo simulations to solve many-body hamiltonians relying on a c++ library (SimpleMC Thomas Hahn). Python codes are used to post-process data, plot visualization and fitting.

To create a python environment with the necessary packages for data analysis type from your login directory:

```
cd DIAGMC-LAB
source ~/DIAGMC-LAB/env_setup
```

This will create a folder `DIAGMC-PY` with the numerical and graphical packages needed for python codes.

Content of `env_setup` file:

```
# python environment for post-processing

env=~/DIAGMC-PY
python3 -m venv $env
source $env/bin/activate
pip install --upgrade pip
pip install numpy matplotlib scipy
```

4. How to run and control a job

The cluster OPH is an environment shared between users of physics from UniBo, both students and researchers (material, theoretical, applied, atmosphere physics and so on).
More info at `https://apps.difa.unibo.it/wiki/oph:cluster`.
There is an interface where users submit tasks handled by a scheduler `SLURM`. The scheduler put the users requests in a queue and then dispatches them to the available compute nodes (order and time of execution depends on the amount of requested resources, see section 6).
We will name each task as a `JOB`.
For each day of activities we will use a job named `blade_run` located in each of the Day# exercises folders (e.g. `~/DIAGMC-LAB/Day1/blade_run`).
That's an example on how to run a Monte Carlo simulation for π estimation during Day1. (again starting from your login directory i.e. `/home/STUDENTI/name.surname00`)

```
cd DIAGMC-LAB
cd Day1
ls          # see list of files and folders
cp blade_run mc_pi
cd mc_pi
sbatch blade_run
```

Before submitting a job we should check the content of the jobscript `blade_run` and eventually modify it when necessary (see section 6).
You can use an editor such as nano or vim.
Important commands:

```
- sbatch: submit job (i.e. sbatch blade_run)
- squeue: check status (squeue -u username)
- scancel: cancel a job (scancel #jobnumber)
```

More info can be found on the wiki `https://apps.difa.unibo.it/wiki/oph:cluster:jobs`

5. **DIAGMC-LAB folder**
   Each exercise folder contains an executable with the same and a `test` subfolder where the exercise will be done. Additional python scripts for post-processing are already inside `test`.

```
DIAGMC-LAB/
|-- Day1
|   |-- blade_run
|   |-- mc_pi
|   |   |-- area_method.py
|   |   |-- histogram.py
|   |   |-- mc_estimator.py
|   |   `-- plot.py
|   `-- mcmc
|       |-- bimodal.py
|       `-- bimodal_2.py
|-- Day2
|   |-- blade_run
|   |-- free_propagator
|   |   |-- free_propagator
|   |   `-- test
|   |       |-- plot_convergence.py
|   |       `-- plot_histogram.py
|   |-- potential
|   |   |-- potential
|   |   `-- test
|   |       `-- plot_fit.py
|   `-- two_level_system
|       |-- test
|       |   `-- post_processing.py
|       `-- two_level_system
|-- Day3
|   |-- blade_run
|   `-- froehlich
|       |-- froehlich
|       `-- test
|           `-- post_processing.py
|-- HOWTO_Day2-3
`-- env_setup
```

   At the root there are the three days activities folders (Day1, Day2, Day3) and the `env_setup` script previously discussed. Day by day we will discuss the structure of each subfolder with the exercises.

6. **Setup a job** The job execution is controlled via the `blade_run` script.
   We consider the `blade_run` job used on Day1.
   The one used for Day2-3 will be looked together in the following lab days.

```
1  #!/bin/bash
2  #SBATCH --constraint=blade    ## only blade nodes
3  #SBATCH --nodes=1
4  #SBATCH --tasks-per-node=1
5  #SBATCH --mem-per-cpu=1G
6  #SBATCH --time=00:05:00
7  #SBATCH --output=a_%N-%j.out
8
9  source ~/DIAGMC-PY/bin/activate
10 export UCX_TLS=ud,sm,self
11
12 python3 area_method.py --N 50000      # area method Nc/N
13 python3 mc_estimator.py --N 50000     # sample from integral
14 python3 plot.py                       # plot convergence for one or both methods
15 python3 histogram.py --N 50000 --M 500  # repeated measurements for both methods
```

1-7: `#SBATCH --option` define resources selection (e.g. partition. number of cpus and memory).
9-10: activate python environment and variables for communication between processors.
12-15: python scripts with different tasks

7. **SSH with graphic support**: The following `ssh` command adds options to avoid disconnections from the server and enable the graphic server to display results.

```
ssh -o ServerAliveInterval=60 -X -J name.surname00@bastion-nav.difa.unibo.it name.
  surname00@ophfe1
```

The command is prompted in a single line. Breakline in the box is only for visualization.

8. **Display images on the cluster**: After enabling graphic support (section 7) you can display images located on the cluster directly on your screen.

```
xdg-open my_image.pdf
```

This command will open a graphical editor available on the cluster and show you the image.

9. **Copying data from cluster to laptop**: use the scp command from your laptop to copy the files you want to store.

```
scp -J name.surname00@bastion-nav.difa.unibo.it name.surname00@ophfe1:/home/STUDENTI/
    name.surname00/myfile.txt .
```

This guide shows the necessary steps to do Monte Carlo calculations in a cluster.
Further informations and details on how to modify the jobs, automatize the calculations will be provided during the lab.